ADRIAN LLOPART, PhD

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Product-oriented Senior AI and Robotics Engineer and founder with a proven track record of building and shipping cutting-edge deep learning models on-cloud and on-device. Passionate about the intersection between AI and Robotics: multimodal perception, VLA policies and embodied AI.

PROFESSIONAL EXPERIENCE

Founder — *Copenhagen, Denmark*
OpenRAL — *Jul 2026 - Present*

- Created **OpenRAL**, an open-source, ROS 2-native **Robot Agentic Layer**: the typed, traceable, safety-first runtime that holds an embodied-AI stack together — fast VLA policies (30-200 Hz), slow LLM reasoning, perception and classical control (github.com/OpenRAL/openral).
- Designed typed Pydantic v2 contracts between layers, hot-swappable robot skills (VLA policies, perception detectors, scene VLMs) and tf2-aware scene graphs for world-state modelling.
- Built a C++ safety kernel with deny-by-default access control and E-stop, with full observability through OpenTelemetry traces replayable in Foxglove.
- Growing an Apache 2.0 community in collaboration with MakerMods and contributors from NYU Shanghai.

Senior AI Engineer (Consultant) — *Copenhagen, Denmark*
Qualia — *2026 - Present*

- Built an egocentric-video-to-robot retargeting pipeline that converts monocular human video into robot training data for VLA policies, integrating 2D hand detection, segmentation, depth estimation, 3D hand prediction and motion retargeting.
- Trained robotic foundation models on the retargeted data and deployed the resulting policies on OpenArms and Flexiv robot arms.

CEO / co-Founder — *Copenhagen, Denmark*
MederiAI — *Feb 2024 - 2026*

- Led development of a spatio-temporal vision AI platform that detects gastrointestinal lesions in video-capsule endoscopy footage, cutting clinician review time from ~2 hours to ~15 minutes while lowering missed-lesion rates.
- Raised 300k€ in pre-seed funding; led a team of full-stack engineers across UI/UX, frontend, backend, AWS infrastructure and AI pipelines.

Senior Machine Learning Engineer — *Copenhagen, Denmark*
Jabra GN — *Aug 2024 - Aug 2026*

- Deployed multimodal deep learning models for real-time visual tracking and speech diarization on edge devices powering videobars for meeting rooms.
- Built the team's MLOps stack from scratch: data versioning, model training orchestration, deployment and monitoring.
- Built the generative Video + Audio data pipeline for synthetic multimodal data generation and annotation.

Senior Machine Learning Researcher *Copenhagen, Denmark*
Veo Technologies *Jan 2021 - May 2024*

- Shipped production deep learning models to cloud and edge devices for automated sports capture: 2D/3D player and ball detection and tracking from monocular video, plus action recognition and localization.
- Owned the full model lifecycle — data collection, annotation, architecture design, training, optimization and deployment — using PyTorch, TensorRT, AWS and GStreamer.
- Co-developed AI-driven advanced match analytics, including the UX/UI of the user-facing product.

Senior AI Algorithms Developer *London, United Kingdom*
Huawei *Jan 2019 - Jan 2021*

- Re-implemented and improved state-of-the-art deep learning methods in TensorFlow and PyTorch for object recognition, detection and segmentation, and human action recognition and detection.
- Developed LiftFormer, a transformer-based 3D human pose estimation model achieving state-of-the-art results on Human3.6M and HumanEva-I with fewer parameters than prior methods.
- Supervised intern research projects.

Supervisor of Master Theses *Lyngby, Denmark*
Technical University of Denmark (DTU) *Feb 2017/18 - Jul 2017/18*

- Supervised five MSc theses on robot motion planning, mobile-robot obstacle avoidance with RNNs and CNN transfer learning; several were presented at international conferences.

PUBLICATIONS

- **Liftformer: 3D human pose estimation using attention models**, ArXiv preprint: *arXiv:2009.00348*, September 2020, UK.
- **Online Semantic Segmentation and Manipulation of Objects in Task Intelligence for Service Robots**, International Conference on Control, Automation, Robotics and Vision (ICARCV), November 18, 2018, Singapore.
- **Semantic mapping and object detection for indoor mobile robots**, Industrial Conference on Robotics and Computer Vision (ICRCV), November 17, 2018, Thailand.
- **Bayesian Convolutional Neural Networks with Variational Inference**, ArXiv preprint: *arXiv:1806.05978v5*, November 14, 2018, Denmark.
- **Outlook for navigation-comparing human performance with a robotic solution**, International Conference on Maritime Autonomous Surface Ships (ICMASS), November 8, 2018, South Korea.
- **A Rule-Based Approach for Constrained Motion Control of a Teleoperated Robot Arm in a Dynamic Environment**, The International Conference on Robotics Systems and Automation Engineering (RSAE), October 6, 2018, Spain.
- **Autonomous 3D model generation of unknown objects for dual-manipulator humanoid robots**, IEEE 5th International Conference on Robot Intelligence Technology and Applications (RITA), Dec 14, 2017, South Korea.
- **Generalized Framework for the Parallel Semantic Segmentation of Multiple Objects and Posterior Manipulation**, IEEE International Conference on Robotics and Biomimetics (ROBIO), Dec 6, 2017, Macao.
- **Door and Cabinet Recognition Using Convolutional Neural Nets and Real-time Method Fusion for Handle Detection and Grasping**, IEEE 3rd International Conference on Control, Automation and Robotics (ICCAR), Apr 22, 2017, Japan.

EDUCATION

PhD in Robotics and Artificial Intelligence — *Lyngby, Denmark*
Technical University of Denmark (DTU) — *Dec 2015 - Dec 2018*
Automation and Control (AUT) Group
Supervisors: Ole Ravn & Nils A. Andersen

PhD External stay — *Daejeon, South Korea*
Korea Advanced Institute of Science and Technology (KAIST) — *Jan 2017 - Sept 2017*
Robot Intelligence and Technology (RIT) Lab
Supervisor: Jong-Hwan Kim

Master thesis: Teleoperation of Miniaturized Humanoid Robots — *Tokyo, Japan*
University of Tokyo (TU) — *April 2015 - Aug 2015*
User Interface Research Group (Igarashi Lab)
Supervisors: Takeo Igarashi & Daisuke Sakamoto

Master of Science in Automation and Robotics — *Lyngby, Denmark*
Technical University of Denmark (DTU) - Double degree programme — *Sept 2013 - Aug 2015*
Study line: Automation and Robotics

Bachelor and Master of Science in Industrial Engineering — *Barcelona, Spain*
Polytechnical University of Catalonia (UPC) — *Sept 2009 - Aug 2013*
Study line: Automation and Robotics

TECHNICAL SKILLS

AI	PyTorch, TensorFlow, TensorRT, Docker, AWS (ECR, EC2, S3 …), Azure, Kubernetes, NeptuneAI, MLFlow, Encord, HuggingFace
Robotics	ROS/ROS 2, OpenCV, PCL, MoveIt, Open3D, VLA policies, tf2
Others	Linux, Python, C/C++, GStreamer, GitHub, Elastic, Notion

LINGUISTIC SKILLS

English	Full Professional Proficiency (C2)
Spanish	Native (C2)
Catalan	Native (C2)
German	Conversational (B2)
Danish	Beginner (A2)